

EXECUTIVE SUMMARY

System Capacity & Care Load Analytics for Unaccompanied Children

The effective management of healthcare capacity is essential for ensuring timely and appropriate care delivery to vulnerable populations. Unaccompanied Children (UC) entering the United States are temporarily placed under the care of the U.S. Department of Health and Human Services (HHS), where continuous monitoring of healthcare demand, care-load patterns, and operational capacity is critical for informed decision-making. Variations in healthcare demand and resource utilization highlight the need for data-driven approaches that support proactive healthcare capacity planning.

This project presents a healthcare analytics framework designed to monitor system capacity and care-load trends among Unaccompanied Children using historical healthcare data spanning January 2023 to December 2025. The proposed framework integrates exploratory data analysis, key performance indicator (KPI) assessment, time-series forecasting, and interactive data visualization techniques to support evidence-based healthcare capacity management.

The analytical findings indicate substantial temporal variations in healthcare demand across the study period. Trend analysis revealed changing care-load patterns and highlighted the importance of continuous operational monitoring. Time-series decomposition identified both long-term trends and seasonal variations within the historical dataset, demonstrating the dynamic nature of healthcare capacity requirements. Furthermore, a 90-day forecasting model was developed to estimate short-term healthcare demand patterns, enabling stakeholders to anticipate resource requirements and improve operational preparedness.

The integration of forecasting techniques with healthcare analytics provides meaningful support for proactive capacity planning and resource allocation decisions. By identifying emerging demand patterns and monitoring key healthcare indicators, the proposed framework contributes to improved operational visibility and facilitates data-driven decision-making processes.

Based on the analytical findings, continuous monitoring of healthcare capacity indicators, the adoption of forecasting-driven planning strategies, and timely resource allocation are recommended to strengthen healthcare system preparedness. These approaches can support more effective responses to fluctuating healthcare demand while improving operational efficiency in healthcare capacity management.

In conclusion, this project demonstrates the value of combining healthcare analytics, forecasting methodologies, and interactive visualization techniques to support evidence-based healthcare capacity planning for vulnerable populations. The proposed framework provides actionable insights that can contribute to informed operational decision-making and enhanced preparedness within healthcare systems managing Unaccompanied Children.